Vidyavardhini's College of Engineering & Technology

Department of Computer Science Engineering (Data Science)

Academic Year: 2026-27

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2026-27

Course: Wireless and Mobile Communication **Course code:**

Year: TE **SEM: V**

Experiment No. 8
Aim: To study path loss in an indoor environment by measuring Wi-Fi signal strength (RSSI) under different obstacle conditions.
Name: Meet Mehta
Roll Number: 30
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	5	3	2
Understanding	5	3	2
Journal work and timely submission.	10	8	4

Checked by

Name of Faculty : Mrs. Sonali Sable

Signature **:** **Date :**

Aim

To study path loss in an indoor environment by measuring Wi-Fi signal strength (RSSI) under different obstacle conditions.

Theory

1. Path Loss

Path loss is the reduction in the power or strength of a wireless signal as it travels from the transmitter to the receiver.

In a wireless communication system, the signal becomes weaker as it travels through the environment due to distance, walls, obstacles, reflection, diffraction, and absorption.

Path loss is an important factor in determining the coverage and reliability of a wireless communication system.

2. RSSI (Received Signal Strength Indicator)

RSSI is a measure of the received signal strength at the receiver. I

For Wi-Fi networks, RSSI is commonly represented in **dBm**.

RSSI values are generally negative. A value closer to zero indicates a stronger received signal.

For example:

- **−40 dBm** → Strong signal
- **−60 dBm** → Moderate signal
- **−75 dBm** → Weak signal

The exact signal-quality interpretation may vary depending on the device and environment.

3. Indoor Path Loss

In an indoor environment, wireless signals may pass through walls, doors, glass, furniture, and other obstacles.

These obstacles can absorb, reflect, or scatter the wireless signal, resulting in a reduction in received signal strength.

Therefore, indoor path loss depends not only on distance but also on the surrounding physical environment.

4. Effect of Obstacles

Different obstacles affect wireless signals differently.

Common indoor obstacles include:

- Walls
- Wooden doors
- Glass doors/windows
- Furniture
- Metal objects
- Multiple walls

A thicker or denser obstacle may cause greater signal attenuation.

5. Effect of Distance

As the distance between the Wi-Fi router and smartphone increases, the received signal generally becomes weaker.

Therefore, RSSI usually becomes more negative as the distance increases.

For example:

Short distance → stronger RSSI

Long distance → weaker RSSI

6. Signal Attenuation

Signal attenuation refers to the reduction in signal strength as the wireless signal travels through a medium or encounters an obstacle.

In an indoor environment, attenuation can occur due to:

- Walls and doors
- Furniture
- Human bodies
- Building materials
- Absorption
- Reflection
- Diffraction

Higher attenuation results in weaker received signal strength.

7. Relationship Between RSSI and Path Loss

RSSI can be used to observe changes in received signal strength under different indoor conditions.

When an obstacle is introduced between the transmitter and receiver:

Obstacle → Signal attenuation → Lower RSSI → Higher path loss

Therefore, comparing RSSI values under different conditions helps us understand the effect of obstacles on wireless signal propagation.

Procedure

Step 1: Select a Wi-Fi Router

Take a Wi-Fi router/access point and place it at a fixed location in the laboratory or classroom.

Keep the router switched ON throughout the experiment.

Step 2: Select a Smartphone

Take a smartphone having Wi-Fi capability.

Install and open a suitable **Wi-Fi Analyzer / Wi-Fi signal monitoring application** that can display the RSSI value of the connected Wi-Fi network.

Step 3: Connect the Smartphone to Wi-Fi

Connect the smartphone to the Wi-Fi network provided by the selected router.

Verify that the smartphone is successfully connected to the router.

Step 4: Fix the Measurement Distance

Select a suitable fixed distance between the router and smartphone, for example **5 m**.

Keep the approximate distance constant while comparing different obstacle conditions.

Step 5: Measure RSSI in Open Space

Place the smartphone such that there is no major obstacle between the router and smartphone.

Observe the RSSI value displayed by the Wi-Fi Analyzer application.

Record the measured RSSI value in the observation table.

Step 6: Introduce the First Obstacle

Place an obstacle between the router and smartphone, such as a **glass door/window**.

Keep the router, smartphone, and measurement distance approximately unchanged.

Step 7: Measure RSSI with the Obstacle

Observe the RSSI value with the obstacle present.

Record the RSSI value in the observation table.

Step 8: Test Different Obstacles

Repeat the experiment using different indoor obstacles, such as:

- Wooden door
- Glass door/window
- One wall
- Two walls
- Furniture, if applicable

Step 9: Record RSSI Values

For each obstacle condition, record the corresponding RSSI value.

If possible, take **2–3 readings** for each condition and calculate the average RSSI.

Step 10: Compare RSSI Values

Compare the RSSI values obtained under:

- Open-space condition
- Single-obstacle condition
- Different obstacle conditions
- Multiple-wall condition

Observe how the received signal strength changes.

Step 11: Perform Distance Testing

If required, repeat the experiment by changing the distance between the router and smartphone.

Suggested distances may be:

1 m → 3 m → 5 m → 7 m → 10 m

Record the RSSI value at each distance.

Step 12: Maintain Similar Experimental Conditions

During distance testing, try to keep:

- Router position fixed
- Smartphone orientation approximately fixed
- Same Wi-Fi network
- Similar environmental conditions

This helps in obtaining more consistent observations.

Step 13: Calculate Average RSSI

If multiple readings are taken at the same condition, calculate the average RSSI.

For example:

Average RSSI =
$(RSSI_1 + RSSI_2 + RSSI_3) / 3$

Step 14: Plot the Graph

Plot a graph of:

Obstacle Condition vs. RSSI

or, for distance testing:

Distance vs. RSSI

Observe the variation in signal strength.

Step 15: Analyze the Results

Compare the RSSI values obtained for different obstacles and distances.

Observe which conditions result in stronger or weaker received signals.

Step 16: Determine the Effect of Obstacles

Identify how walls, doors, glass, and other obstacles affect the received Wi-Fi signal.

Generally, an increase in obstacles causes greater attenuation and a weaker received signal.

Step 17: Record Observations and Conclusion

Record all experimental observations in the observation tables.

Write the conclusion based on the measured RSSI values and the effect of obstacles on wireless signal propagation.

OUTPUT

Observation Table – RSSI vs. Obstacles

Sr. No.	Obstacle / Condition	Distance (m)	RSSI (dBm)	Signal Observation
1	Open space	5	-13	Strong signal
2	Glass door/window	5	-31	Strong signal
3	Wooden door	5	-53	Moderate signal
4	One wall		-61	Moderate signal
5	Two walls	5	-78	Weak signal
6	Other obstacle	5	99	Weak signal

Note: The values in the table should be filled using the actual experimental measurements. The distance and obstacles shown are suggested test conditions and may be modified according to the laboratory environment.

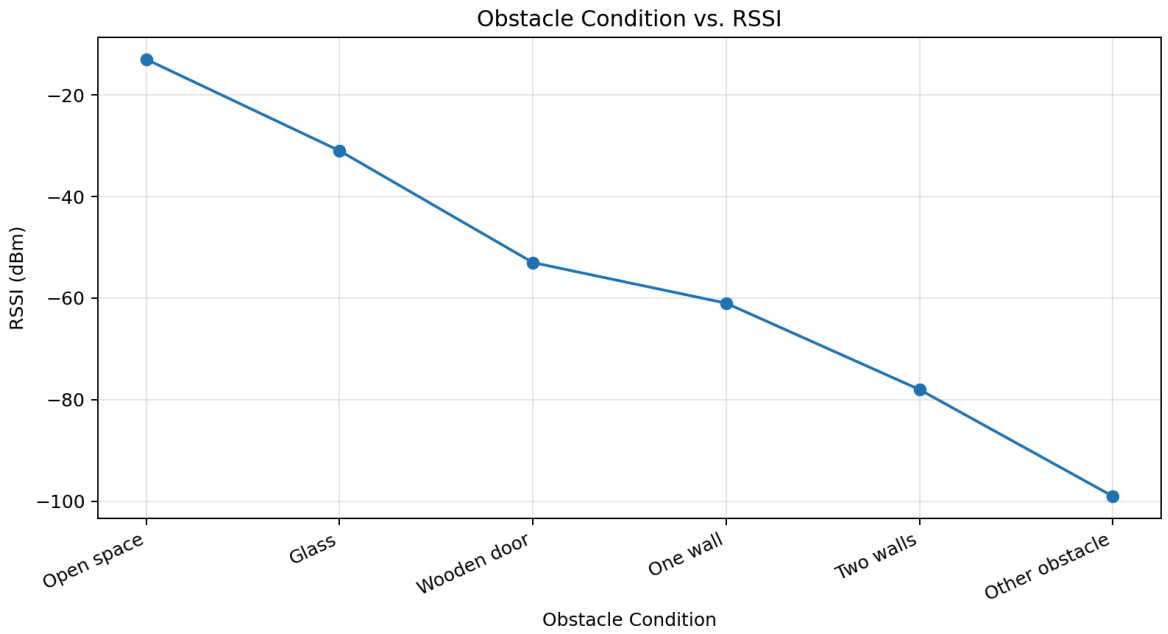
Observation Table – RSSI vs. Distance

Sr. No.	Distance (m)	RSSI (dBm)	Signal Observation
1	1	-13	Strong
2	3	-20	Strong
3	5	-31	Strong

4	7	-43	Strong
5	10	-55	Moderate

Note: The above distances are suggested test points. The actual measurable range depends on the Wi-Fi router, smartphone, indoor environment, and surrounding interference.

Graph



Graph 1: Obstacle Condition vs. RSSI

- X-axis: **Obstacle Condition**
- Y-axis: **RSSI (dBm)**

Graph 2: Distance vs. RSSI

- X-axis: **Distance (m)**
- Y-axis: **RSSI (dBm)**

Path Loss Analysis

Condition	RSSI Observation	Path Loss / Attenuation Observation
Open space	-13 dBm	Very low attenuation; strongest signal.
Glass obstacle	-31 dBm	Low attenuation; signal remains strong.
Wooden door	-53 dBm	Moderate attenuation; signal becomes weaker.
One wall	-61 dBm	Higher attenuation; moderate/weak signal.
Multiple walls	-78 dBm	High attenuation; weak signal.

General Observation

- Higher RSSI (closer to 0 dBm) indicates a stronger received signal.
- Lower RSSI (more negative value) indicates a weaker received signal.
- Obstacles generally reduce the received signal strength.
- Increasing the distance generally reduces the received signal strength.
- Different obstacles produce different amounts of signal attenuation.

Range / Signal Analysis

Condition	Signal Strength	Observation
Open space	Strong	
Single obstacle	Moderate	
Multiple obstacles	Weak	

Maximum observed attenuation	86 dB (from -13 dBm to -99 dBm)	

Conclusion:

The experiment successfully demonstrated the effect of obstacles and distance on Wi-Fi signal strength in an indoor environment. The measured RSSI values showed that the signal was strongest in open space and became weaker when obstacles such as glass, wooden doors, and walls were introduced. Multiple walls produced the greatest attenuation among the tested conditions. It was also observed that increasing the distance between the router and smartphone generally caused RSSI to become more negative. Thus, indoor obstacles and distance contribute to path loss and attenuation, resulting in reduced received Wi-Fi signal strength.

Viva Questions

1. What is path loss in wireless communication?
   Path loss is the reduction in the power or strength of a wireless signal as it travels from the transmitter to the receiver due to distance, walls, obstacles, reflection, diffraction, and absorption.

2. What is RSSI?
   RSSI (Received Signal Strength Indicator) is a measure of the received wireless signal strength at the receiver. It is commonly represented in dBm for Wi-Fi.

3. Why is RSSI generally represented using negative dBm values?
   Received Wi-Fi power is usually much lower than 1 mW, so its value in dBm is negative. A value closer to 0 dBm represents a stronger signal.

4. What happens to RSSI when the distance increases?
   RSSI generally becomes more negative, meaning the received signal becomes weaker.

5. How do walls and other obstacles affect wireless signals?
   Walls and obstacles absorb, reflect, or scatter the signal, causing attenuation and reducing the received signal strength.

6. What is signal attenuation?
   Signal attenuation is the reduction in signal strength as a wireless signal travels through a medium or encounters an obstacle.

7. What is the difference between RSSI and path loss?
   RSSI indicates the signal strength received by the receiver, whereas path loss represents the reduction in signal power between the transmitter and receiver.

8. Why should the router position be kept fixed?
   The router should remain fixed so that changes in RSSI are mainly due to the tested obstacles or distance rather than changes in the transmitter position.

9. Why should the smartphone orientation be kept approximately constant?
   Smartphone orientation can affect the received signal. Keeping it approximately constant makes the measurements more consistent.

10. Which generally gives a stronger signal: open space or multiple walls?
    Open space generally gives a stronger signal because there are fewer obstacles and less attenuation.

11. What is the relationship between distance and received signal strength?
    As distance increases, received signal strength generally decreases, so RSSI becomes more negative.

12. Name some common indoor obstacles that cause signal attenuation.
    Walls, wooden doors, glass doors/windows, furniture, metal objects, and multiple walls.

13. Why can two different obstacles produce different RSSI values?
    Different materials have different absorption, reflection, and scattering characteristics, so they cause different amounts of attenuation.

14. What factors other than obstacles can affect Wi-Fi signal strength?
    Distance, reflection, diffraction, absorption, building materials, furniture, human bodies, and surrounding interference can affect Wi-Fi signal strength.

15. What happens to wireless communication when path loss increases?
    The received signal becomes weaker, which can reduce communication reliability and coverage.